OPERATIONS PLAN - UAV SPRAYING
ES0023781

Task description : Twins 317 holding spray drone	Total ha	60.81

HQPlantations responsible officer	**Liam Browne**	**Task stage**	**Approved**
Contractor	Nat. Drones	**Purchase order no.**	043186
Start date	02/03/2026	**Finish date**	

Asset details

Location/Stand	Net (treated) ha
589_TWINS_317 Design	60.81
Total treated ha	**60.81**

Task specification

Application volume (L mix/ha)	50.00
Chemical mix and application rate (product per ha)	Glymac 450 @ 4L/ha ;;Starane @ 1.8L/ha ;;Endorse @ 1L/100L
Preferred wind direction	SW

Additional specifications
Holding spray. Some standing trees/stags, approval to fly at 8m if needed. Was Twins 217, now third rotation DO NOT enter into other's worksite without permission, follow UHF protocol to communicate. Minimum Requirements - provide the following - The contractor must operate within the parameters, requirements and guidelines defined in their Service Agreement. - The contractor must provide HQPlantations with operational records (daily work progress - hard copy/electronic) as defined in their Service Agreement. Records must be provided in timely manner and in a appropriate format. - Complete 'Site Awareness Checklist' before work commences - Note any operational hazards and report any Incidents immediately to HQPlantations - Follow emergency and location Map1 for emergency contacts detail.

Quality management requirements

Prior to commencing the task, the contractor must clearly mark on the operations map (Map 3) the required downwind buffer distances based on prevailing weather conditions and the risk to adjoining areas, internal watercourses and retained vegetation.

As soon as practicable after completing the operation, the contractor must give HQPlantations:

(a) a copy of the operations map (Map 3) marked up to show the planned downwind buffer areas and any areas excluded due to presence of surface water or other reasons;
(b) the completed **aerial spraying application record** and **contractor self assessment QA checklist**;
(c) a shapefile detailing the spray flight path for the task.

Operations - Quality Assurance
All contractors required to complete supplied QAs during operation, QAs will need to be submitted before payment

Known worksite hazards and special management requirements

SAFETY ISSUES
Fire Protection Smoking is only permitted on bare earth at least 2m away from any flammable material. Butts to be disposed of in a suitable container. During high fire dangerHQP may prohibit smoking on HQP lands for a specified period. Hot Works, Request permission from HQP Supervisor (via text or phone) prior to undertaking any hot works on site and comply with any fire prevention conditions imposed by HQP. Fire suppression must be available on site. **General** Stream bank drop offs - Are associated with all gullies, proceed near gullies with caution, square on, boom as low to the ground as possible, clear undergrowthto assess ground conditions. Rubbish - May be present on site. DO NOT TOUCH.Call HQPlantations supervisor to organise removal. Standing dead trees withincompartment. Machine operators and staff to be aware of surroundings when working around dead trees **Public Use** Contractors to note that Beerburrum-Woodford Rd and Old Gympie Rd have high public use by agricultural, recreational vehicles and motorbikes/ general public.Contractor to be aware that the public regularly access this area. Ensure signage is in place and cease work immediately if an unauthorised person enters your site. Report all occurrences to HQP staff. Safety barriers may be utilised tocontrol site access, this may need to be adjusted as required. Restrictions on weekend work may apply, check with HQP before conducting any weekend/after hours operationalactivities. **Signage** Contractors are to erect worksite safety singage as per HQP 'Signage Guidelinesfor Forest Operations'

ENVIRONMENTAL ISSUES

Conservation
No machinery movement permitted within native forest areas. Wildling removal ispermitted if within the operational area on HQP managed lands.

Native retained vegetation around the waterways in this area could form part of the Wallum Froglet and Tusked Frog habitat. Management processes are found in the EVNT booklet

Koalas are known to occur within this area. Please take additional care when driving to and from the Operations Area and report any sightings to the HQP Representative. If a koala is found injured please call the RSPCA call centre on 1300264625. Refer to HQP Koala Management Operations Guidelines.

Declared pests/weeds
Wash down machinery prior to entry to site and before leaving compartment. Inform HQP of any infestations of pests observed in the operational area and on the access route.

Refueling
Refueling must only occur over 50m from waterways in an area where any spills will not effect private property, gazetted roads, watercourses or native retainedareas.
Maintain spill kits on site. Report all incidents.

Water Course Protection
Maintain WPZ as specified on OPs map. No machinery movements permitted within these areas unless authorised by HQP supervisor.

OPERATIONAL ISSUES

Cultural Heritage
Glass House Mountains National Park is situated across the southern and easterncompartment road. This is an area of significant cultural heritage and conservation value. Do not enter this area with machinery.

Nil First Nations and Nil modern recorded in online databases or found during site inspection.
If any object is found that could be of cultural or historical importance please inform HQP supervisor.

Foliage Sale Area
Be aware this compartment in within a foliage sales area agreement, contractor to be aware there may be personel on site

Powerlines and Utilities
Contractor to be aware there is powerlines running along Old Gympie Rd, to the east and outside of operational area.

Exclusion Zone
There is a severely washed out gully in the centre of the compartment, this gully has unstable trees in it. Remain more than two tree lengths away from this gully. Do not enter exclusion zone

Bridge
Bridge over creek crossing in East of compartment not suitable for heavy machinery or trucks to cross. Only light vehicles. Marked on map

UAV Spraying
Isolated tree spars may be present in operational area, contractor is responsible for undertaking initial fly over and risk assessment of the site to determineappropriate controls for these & other site/nearby hazards.

Contractor is responsible for supplying flight path used throughout the operations (shapefile). The data should be submitted to HQP Supervisor prior to invoicing.

Batteries checked for health and charging cycles before use. Chemical tank cleaned to prevent contamination. Flight planning software updated.

Fields surveyed and mapped prior to application. Restricted zones marked (waterways, sensitive crops). Boundaries uploaded into drone system.

Chemcial type, concentration, and load recorded per flight. Weather conditions logged (wind speed, humidity, temperature). Flight path, coverage area, and application rate documented. Nozzle settingsadjusted if drift observed.

Operations representatives and delegates

Company	Name	Phone	Mobile	Radio

Operations plan consultation, induction and approval record

By agreeing to this operations plan, the Contractor agrees that they:

1. have been consulted on the operations plan and are satisfied they are able to safely and efficiently complete the task in accordance with the plan and applicable agreement;
2. have received maps detailing access, evacuation points, treatment areas, known hazards and special management areas;
3. have received relevant forms, working maps and other documentation required to complete the task (eg. QA form, incident report form, working maps, SDS for any chemical supplied by HQPlantations);
4. have been advised by HQPlantations of known site hazards and have had an opportunity to provide input into the controls to eliminate these hazards;
5. will conduct their own site hazard identification and risk assessment, through the implementation of the contractor's own health and safety management system;
6. will advise HQPlantations of any new hazards or issues that require variations to this plan, as they arise; and
7. will ensure that all persons invited onto the site (including the contractor's workers) have completed a HQPlantations "Plantation Safety Induction".

Company	Operations representatives name	Signature	Date
HQPlantations	Cameron Frame	C.J Frame	02/03/2026
Contractor			

Operations plan amendments following contractor consultation

Amendment details	Contractor representative (name and sign)	HQPlantations representative (name and sign)	Date

Operations completion record

Status*	Location of completed part area/s	Contractor representative (name and sign)	HQPlantations representative (name and sign)	Date

*Status options (Full completion, Partial completion, Re-commencement)

Aerial Spraying Application Record (to be completed by contractor)

Start Date		**Start Time**		**Finish Time**	
Target Area ha			**Estimated Treated Area ha**		

UAV Details

Pilot in Charge		**UAV Make/Model**	
Registration		**Nozzle Type & Code No.**	
Av. Flying Height	m	**Flying Speed**	km/h
Swath Width	m	**Application Rate**	L/ha
Application Method (tick applicable)		**Single Pass** ☐	**Double Pass** ☐

Expected Weather

Weather Conditions - Minimum Hourly, or as soon as conditions change

Time							
Temp °C							
RH %							
Delta T							
Wind Direction							
Wind Speed km/hr							
Major Risk Buffer (m)							
Moderate Risk Buffer (m)							

Product & Application Rates Used

Product					
Target Rate (L/ha)					
Target Volume (L)					
Actual Volume (L)					
Actual Rate (L/ha)					

HQP to complete (product stocktake)

Volume product delivered to site					
Volume product unused (to return)					
Calculated volume used					

Completed by (contractor)		**Signature**	
Checked by (HQP)		**Signature**	

Determining minimum DOWNWIND Buffer Distances

1. When using CDA nozzles with droplet size >550 micrometre VMD

MINIMUM DOWNWIND BUFFER DISTANCES(1)			
Wind Speed (km/hr)	**Moderate Risk (m)**	**Major Risk (m)**	**Severe Risk (m)**
0 to <3	No Spraying		
3 to <8	2	5	Do not apply when risk area downwind of application (2)
8 to <14	5	10	

2. When using nozzles with minimum Coarse VMD (326 - 400 micrometre)

MINIMUM DOWNWIND BUFFER DISTANCES(1)			
Wind Speed (km/hr)	**Moderate Risk (m)**	**Major Risk (m)**	**Severe Risk (m)**
0 to <3	No Spraying		
3 to <8	10	15	Do not apply when risk area downwind of application (2)
8 to <14	15	30	

Additional notes:

(1) Buffers are all based on a release height of 4-6m. There is allowance to release from up to 10m to avoid flying hazards, noting that minimum no-spray safety buffers must be doubled.

(2) Severe risk areas requiring a 300m helicopter exclusion can be treated by drone only when there is positive wind direction away from the risk area.

All UAV applications of liquid herbicide are to contain a spray oil to mitigate production of fine droplets.

HQP drainage line classification and aerial spray buffer requirements

HQP maintains three different drainage line layers, each derived form different sources. The following guide outlines how each should be buffered from aerial spray operations.

1. Drainage Field Checked

HQP have assessed the available information including site inspection and have classified drainage lines and assigned a WPZ set back (e.g. plantation excluded from 2m of edge of drainage line) to encourage natural vegetation and bank stabilisation. For drainage lines classified as large or small streams HQPs emphasis is on both the vegetated riparian zone and water quality. For the lower level gullies and waterways, the emphasis is primarily on water protection.

Map symbolisation and spray buffer management:

Drainage Field Checked

Class

Large Stream (WPZ 20m)	Major risk, Buffer off NF edge
Small Stream (WPZ 10m)	
Major Gully (WPZ 5m)	Moderate risk if dry, Major risk if wet, buffer off drainage line
Minor Gully (WPZ 2m)	Minor risk if dry (ok to spray with liquid knockdown), Major risk if wet, buffer off drainage line
Stable Waterway (WPZ 0m)	
Constructed Drain (WPZ 0m)	
Isolated Water Feature	Assess risk (Moderate for internal dam, Major for natural lagoons or external dams, Severe for public water reservoirs)

2. SWAT modelled drainage

This is modelled drainage based off HQPs LiDAR digital elevation model. It is classified into catchment sizes that generally reflect HQPs drainage line classification (see section 3). The modelled data is fairly accurate where there is some slope but is less reliable in floodplain situations or where there has been significant man-made drainage engineering. **Confirm accuracy and drift risk on site with HQP prior to commencing operations.**

Map symbolisation and spray buffer management:

SWAT Catchment Area

300ha Catchment	Major risk, Buffer off NF edge
80ha Catchment	Moderate risk if dry, Major risk if wet, buffer off drainage line
15ha Catchment	Minor risk if dry (ok to spray with liquid knockdown), Major risk if wet, buffer off drainage line
<15ha Catchment	

3. Mapped drainage - unclassified

Uncategorised drainage lines historically drawn from aerial photos. Useful for identifying old man made drains, dams and waterholes but no differentiation between minor gullies and rivers.

Map symbolisation and spray buffer management:

Drainage Unclassified	To be assessed on site and HQP to advise drift risk and buffer application